

Research internship report:

Does data interpolation contradict statistical optimality?

Anas Amrouche

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1 Introduction

When designing a function estimator or statistical classifier, a trade-off is often required to achieve the lowest possible statistical risk, given a dataset $(X_i, Y_i)_{1 \leq i \leq n}$ where the X_i serve as covariates (also referred to as “features” in machine learning) and the Y_i denote the target variables (or “labels”).

Risk is the metric to be optimized and is typically given by $R(\theta, \hat{\theta}) = \mathbb{E}[(\theta - \hat{\theta})^2]$ in the case of quadratic risk. Alternative formulations of the risk may be considered (any L^p norm being suitable); however, the quadratic risk is particularly appealing because it is everywhere differentiable—facilitating gradient-based optimization algorithms—and admits a decomposition

into a bias term and a variance term:

$$R(\theta, \hat{\theta}) = \mathbb{E}[(\theta - \hat{\theta})^2] = (\mathbb{E}[\theta - \hat{\theta}])^2 + \text{Var}(\hat{\theta}).$$

A widely accepted principle in statistical estimation and learning is that there is a trade-off between these two quantities. Increasing the complexity of a model to achieve an increasingly accurate fit to the observed data reduces its bias. However, greater model complexity also increases its sensitivity to the data and, consequently, its variance. This phenomenon, commonly referred to as overfitting in machine learning, can be compared to a human learner who memorizes information without attempting to identify underlying patterns or generalizable structures. This trade-off, which may be reminiscent of an uncertainty principle in quantum mechanics, is in fact only an empirical rule and can be circumvented under certain assumptions.

In the paper by Belkin, Rakhlin, and Tsybakov [1], it is shown that, under a specific interpolation scheme, an estimator can, in certain settings, achieve asymptotic minimax optimality while simultaneously interpolating the training data exactly.

2 The singular Nadaraya–Watson estimator

Definition 2.1 (Nadaraya–Watson estimator)

Let $K : \mathbb{R}^d \rightarrow \mathbb{R}_+$ be a symmetric kernel with its support included in the unit ball (i.e. $K(x) = K(y)$ for all x, y such that $\|x\| = \|y\|$, and $\{x \mid K(x) > 0\} \subseteq B(0, 1)$).

The Nadaraya–Watson estimator of a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ associated with K is defined as follows:

$$\hat{f}_n(x) = \begin{cases} Y_i & \text{if } x = X_i \text{ for some } i \in \{1, \dots, n\}, \\ 0 & \text{if } \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right) = 0, \\ \frac{\sum_{i=1}^n Y_i K\left(\frac{x - X_i}{h}\right)}{\sum_{i=1}^n K\left(\frac{x - X_i}{h}\right)} & \text{otherwise.} \end{cases}$$

This estimator can be seen as a weighted sum over all labels from the training data with the kernel values acting as the weights. The inclusion of the kernel's support in the unit ball prevents data points at a distance greater than h from x (such that $\|x - X_i\| > h$) from influencing the inference. The bandwidth h is one of the hyperparameters of this model.

The **singular Nadaraya–Watson estimator** appears in the special case where K tends to infinity when approaching 0. Since

$$\lim_{x \rightarrow X_j} \frac{\sum_{i=1}^n Y_i K\left(\frac{x - X_i}{h}\right)}{\sum_{i=1}^n K\left(\frac{x - X_i}{h}\right)} = Y_j,$$

the estimator is still continuous at the data points. For the rest of this presentation and the coming simulations, the chosen singular kernel is the one used in the original article:

Definition 2.2 (The kernel K)

$$K(u) = \frac{1}{\|u\|^\alpha} \mathbf{1}(\|u\| \leq 1),$$

with the hyperparameter α satisfying $0 < \alpha < d/2$.

In what follows, we write $K_h(x) = K(x/h)$, and we denote by $\Sigma(\beta, L)$ the space of (β, L) -Hölder functions.

Definition 2.3 (The (β, L) -Hölder function class)

A function f is said to be (β, L) -Hölder if:

- $\forall x, y \in \mathbb{R}^d$, $|f(x) - f(y)| \leq L\|x - y\|^\beta$, if $\beta \in]0, 1]$;
- f is continuously differentiable and $\forall x, y \in \mathbb{R}^d$, $|f(x) - f(y) - \langle \nabla f(y), x - y \rangle| \leq L\|x - y\|^\beta$, if $\beta \in]1, 2]$.

2.1 In the case $\beta \in]0, 1]$ **2.1.1 Statement and assumptions****Theorem 2.1** (Asymptotic optimality in the minimax sense)

Suppose that:

1. $f \in \Sigma(\beta, L_f)$ for $\beta \in]0, 1]$;
2. $\forall x \in \mathbb{R}^d$, $f(x) = \mathbb{E}[Y \mid X = x]$ exists and $\mathbb{E}[(Y - f(x))^2 \mid X = x] \leq \sigma_\xi^2 < +\infty$;
3. the marginal density p of X satisfies on its support $0 < p_{\min} \leq p(x) \leq p_{\max} < +\infty$.

Then, for any fixed x in the support of p ,

$$\mathcal{R}(\hat{f}_n) = \mathbb{E} \left[\left(\hat{f}_n(x) - f(x) \right)^2 \right] \leq C n^{-\frac{2\beta}{2\beta+d}}$$

with C a constant not dependent on the value of n and for $h \sim n^{-\frac{1}{2\beta+d}}$.

Condition 1 is a regularity constraint on the function. It ensures uniform continuity without requiring differentiability, thereby maintaining the utmost generality.

Condition 2 ensures that the variance of the difference between the true value of f and the error-prone measurements Y_i is finite. Consequently, the errors in Y_i remain controllable. Such measurement errors frequently occur in physics, as measuring instruments are never ideal. Therefore, it is commonly assumed that $Y_i = f(X_i) + \xi_i$, where $\xi_i \sim \mathcal{D}$ and $\mathcal{D}$ is a centered probability distribution with finite variance (thus satisfying Condition 2).

The final condition requires the marginal density to be compactly supported and to not tend to 0 at any point. This allows the support of the target function to be fully “covered” by a finite sample of data for all h when n is big enough. If the density were approaching 0 at a point x in its support, the consequence would be that finding points increasingly close to x in the dataset would become more and more difficult as the scope h of the kernel K is getting smaller.

2.1.2 Proof: decomposition of the risk

The proof will rely on estimating $f(0)$ without loss of generality, since the problem remains invariant under a simple translation.

It will consist of upper bounding the loss by three terms:

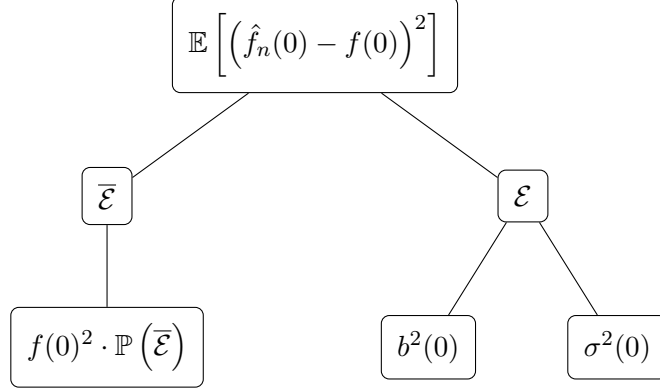


Figure 1: The leaves are the terms bounding the risk.

Let us define the event

$$\mathcal{E} = \left\{ \sum_{i=1}^n K_h(X_i) \neq 0 \right\} = \{\exists i, \|X_i\| \leq h\}$$

where one of the X_i is close enough to 0 for the estimator to compute a value.

Its complement $\bar{\mathcal{E}}$ corresponds to the scenario where the estimator fails to approximate f because the X_i are too far away. On this event, $\hat{f}_n(0)$ vanishes, and hence the squared error will be equal to $f(0)^2$. The probability of this event occurring can be bounded as follows:

$$\begin{aligned} \mathbb{P}(\bar{\mathcal{E}}) &= \mathbb{P}(\forall i \in \{1, \dots, n\}, \|X_i\| > h) = \mathbb{P}\left(\bigcap_{i=1}^n \{\|X_i\| > h\}\right) = \prod_{i=1}^n \mathbb{P}(\|X_i\| > h) \\ &= (1 - \mathbb{P}(\|X_1\| \leq h))^n. \end{aligned}$$

This is where Condition 3 proves useful. It ensures that the probability of the event $\bar{\mathcal{E}}$ vanishes as the sample size increases indefinitely. This translates into the inequality $\mathbb{P}(\|X_1\| \leq h) \geq Cp_{\min}h^d$, where C is a constant independent of n reflecting the equivalence between the $\|\cdot\|_2$ and $\|\cdot\|_\infty$ norms.

Returning to $\bar{\mathcal{E}}$, this yields an exponential decay of the associated probability:

$$\mathbb{P}(\bar{\mathcal{E}}) \leq (1 - Cp_{\min}h^d)^n \leq e^{-nCp_{\min}h^d}.$$

Qualitative analysis. The exponential decay can be understood from the fact that bounding the probability that $\mathcal{E}$ fails to occur as n grows yields a geometric decay in n .

The constant C reflects the equivalence of norms in finite-dimensional spaces.

The lower bound $p_{\min}$ constrains the extent of the marginal density's support: a larger value implies a more concentrated support that is covered more rapidly, thereby accelerating the convergence of the aforementioned bound to 0.

The final term h^d in this exponential reflects the necessity of scaling the radius h of the balls with respect to the dimension d of the feature space. Since h must vanish asymptotically, it will

eventually satisfy $h < 1$. Under this condition, increasing the dimension d decreases h^d , thus requiring a larger bandwidth h in higher dimensions. Nevertheless, for any fixed dimension d , this exponential bound remains asymptotically negligible.

Now that the loss has been bounded on the event $\bar{\mathcal{E}}$, let us focus on the opposite case. On $\mathcal{E}$,

$$\hat{f}_n(0) = \frac{\sum_{i=1}^n Y_i K_h(X_i)}{\sum_{i=1}^n K_h(X_i)}.$$

Proposition 2.1 (Bias–variance decomposition)

$$\begin{aligned} \mathbb{E} \left[\left(\hat{f}_n(0) - f(0) \right)^2 \right] &= \mathbb{E} \left[\left(\hat{f}_n(0) - f(0) \right)^2 \mathbf{1}(\mathcal{E}) \right] + f(0)^2 \mathbb{P}(\bar{\mathcal{E}}) \\ &= \mathbb{E} \left[\left(\hat{f}_n(0) + \mathbb{E} \left[\hat{f}_n(0) \mid X \right] - \mathbb{E} \left[\hat{f}_n(0) \mid X \right] - f(0) \right)^2 \mathbf{1}(\mathcal{E}) \right] + f(0)^2 \mathbb{P}(\bar{\mathcal{E}}) \\ &= \underbrace{\mathbb{E} \left[\left(\hat{f}_n(0) - \mathbb{E}[\hat{f}_n(0) \mid X] \right)^2 \mathbf{1}(\mathcal{E}) \right]}_{\text{Variance term } \sigma^2(0)} + \underbrace{\mathbb{E} \left[\left(\mathbb{E}[\hat{f}_n(0) \mid X] - f(0) \right)^2 \mathbf{1}(\mathcal{E}) \right]}_{\text{Bias term } b^2(0)} + f(0)^2 \mathbb{P}(\bar{\mathcal{E}}) \end{aligned}$$

Here $\mathbb{E}[\cdot \mid X]$ denotes the conditional expectation given the sample $(X_1, \dots, X_n)$, and we used the previous result to obtain the first equality. To get the last equality, implicitly:

$$\begin{aligned} &\mathbb{E} \left[\left(\hat{f}_n(0) + \mathbb{E} \left[\hat{f}_n(0) \mid X \right] - \mathbb{E} \left[\hat{f}_n(0) \mid X \right] - f(0) \right)^2 \mathbf{1}(\mathcal{E}) \right] \\ &= \mathbb{E} \left[\left(\hat{f}_n(0) - \mathbb{E}[\hat{f}_n(0) \mid X] \right)^2 \mathbf{1}(\mathcal{E}) \right] + \mathbb{E} \left[\left(\mathbb{E}[\hat{f}_n(0) \mid X] - f(0) \right)^2 \mathbf{1}(\mathcal{E}) \right] \\ &\quad + 2 \mathbb{E} \left[\left(\hat{f}_n(0) - \mathbb{E}[\hat{f}_n(0) \mid X] \right) \left(\mathbb{E}[\hat{f}_n(0) \mid X] - f(0) \right) \mathbf{1}(\mathcal{E}) \right] \end{aligned}$$

and, since $(\mathbb{E}[\hat{f}_n(0) \mid X] - f(0)) \mathbf{1}(\mathcal{E})$ only depends on $(X_1, \dots, X_n)$, conditioning on the sample gives

$$\begin{aligned} &\mathbb{E} \left[\left(\hat{f}_n(0) - \mathbb{E}[\hat{f}_n(0) \mid X] \right) \left(\mathbb{E}[\hat{f}_n(0) \mid X] - f(0) \right) \mathbf{1}(\mathcal{E}) \right] \\ &= \mathbb{E} \left[\underbrace{\mathbb{E} \left[\hat{f}_n(0) - \mathbb{E}[\hat{f}_n(0) \mid X] \mid X \right]}_{=0} \left(\mathbb{E}[\hat{f}_n(0) \mid X] - f(0) \right) \mathbf{1}(\mathcal{E}) \right] = 0. \end{aligned}$$

The rest of the proof will consist in bounding these two terms by using the hypotheses.

2.1.3 Bounding the variance term

Writing $\xi_i = Y_i - f(X_i)$,

$$\begin{aligned}\sigma^2(0) &= \mathbb{E} \left[\left(\frac{\sum_{i=1}^n \xi_i K_h(X_i)}{\sum_{i=1}^n K_h(X_i)} \right)^2 \mathbf{1}(\mathcal{E}) \right] \leq \sigma_\xi^2 \mathbb{E} \left[\frac{\sum_{i=1}^n K_h(X_i)^2}{\left(\sum_{i=1}^n K_h(X_i) \right)^2} \mathbf{1}(\mathcal{E}) \right] \\ &= \sigma_\xi^2 \cdot n \mathbb{E} \left[\frac{K_h(X_1)^2}{\left(\sum_{i=1}^n K_h(X_i) \right)^2} \mathbf{1}(\mathcal{E}) \right].\end{aligned}$$

The inequality holds because, conditionally on $(X_1, \dots, X_n)$, the ξ_i are independent and centered with conditional variance at most σ_ξ^2 , so that the cross terms $\mathbb{E}[\xi_i \xi_j \mid X]$, $i \neq j$, vanish.

By denoting σ_X^2 the term

$$n \mathbb{E} \left[\frac{K_h(X_1)^2}{\left(\sum_{i=1}^n K_h(X_i) \right)^2} \mathbf{1}(\mathcal{E}) \right]$$

we aim to show that its convergence is polynomial.

Let $\eta_i = \mathbf{1}(\|X_i\| \leq h)$ and $\bar{p} = \mathbb{P}(\|X_1\| \leq h)$, and let us denote

$$\mathcal{A} = \left\{ \sum_{i=1}^n \eta_i \leq \frac{n\bar{p}}{2} \right\}.$$

The bound $\frac{n\bar{p}}{2}$ has not been chosen randomly: it will allow us to use Bernstein's inequality later.

By partitioning the space with $\mathcal{A}$:

$$\sigma_X^2 \leq n \mathbb{E} \left[\frac{K_h(X_1)^2}{\left(\sum_{i=1}^n K_h(X_i) \right)^2} \mathbf{1}(\mathcal{A} \cap \mathcal{E}) \right] + n \mathbb{E} \left[\frac{K_h(X_1)^2}{\left(\sum_{i=1}^n K_h(X_i) \right)^2} \mathbf{1}(\bar{\mathcal{A}} \cap \mathcal{E}) \right] \quad (1)$$

If $\|X_i\| \leq h$, then $K_h(X_i) \geq 1 = \eta_i$; otherwise, $0 = \eta_i \leq K_h(X_i)$. Hence,

$$\sum_{i=1}^n K_h(X_i) \geq \sum_{i=1}^n \eta_i.$$

The second term can be upper bounded by:

$$n \mathbb{E} \left[\frac{4}{(n\bar{p})^2} K_h(X_1)^2 \right].$$

The kernel K belongs to L^2 . By radial symmetry and a spherical change of variables in $\mathbb{R}^d$,

$$\mathbb{E} [K_h(X_1)^2] \leq h^d p_{\max} \int_{\mathbb{R}^d} K^2(u) du \leq h^d p_{\max} C_K \int_0^1 r^{-2\alpha} r^{d-1} dr.$$

Choosing $\alpha < \frac{d}{2}$ ensures that $d - 1 - 2\alpha > -1$, thereby establishing the finiteness of $\mathbb{E} [K_h(X_1)^2]$. Consequently, since $\bar{p} \geq Cp_{\min} h^d$, the second term converges at a rate of $O\left(\frac{1}{nh^d}\right)$.

Returning to the term

$$n \mathbb{E} \left[\frac{K_h(X_1)^2}{\left(\sum_{i=1}^n K_h(X_i)\right)^2} \mathbf{1}(\mathcal{A} \cap \mathcal{E}) \right],$$

it is upper bounded by

$$n \mathbb{P} \left(\sum_{i=1}^n \eta_i \leq \frac{n\bar{p}}{2} \right) \quad \text{since} \quad K_h(X_1)^2 \leq \left(\sum_{i=1}^n K_h(X_i) \right)^2.$$

The event

$$\left\{ \sum_{i=1}^n \eta_i \leq \frac{n\bar{p}}{2} \right\} = \left\{ \sum_{i=1}^n (\eta_i - \bar{p}) \leq -\frac{n\bar{p}}{2} \right\} \quad \text{is included in} \quad \left\{ \left| \sum_{i=1}^n (\eta_i - \bar{p}) \right| \geq \frac{n\bar{p}}{2} \right\},$$

where the variables η_i are independent Bernoulli variables with parameter $\bar{p}$. The probability of this latter event can be bounded using Bernstein's inequality, as the variables $\eta_i - \bar{p}$ are zero-mean and bounded almost surely by 1 in absolute value.

$$\begin{aligned} n \mathbb{P} \left(\sum_{i=1}^n \eta_i \leq \frac{n\bar{p}}{2} \right) &\leq 2n \exp \left(\frac{-\left(\frac{n\bar{p}}{2}\right)^2}{2 \sum_{i=1}^n \mathbb{E}[(\eta_i - \bar{p})^2] + \frac{2}{3} n\bar{p}} \right) = 2n \exp \left(-\frac{n\bar{p}}{8((1 - \bar{p}) + \frac{1}{3})} \right) \\ &\leq 2n \exp \left(-\frac{n\bar{p}}{8 \cdot \frac{4}{3}} \right) = 2n \exp \left(-\frac{3n\bar{p}}{32} \right). \end{aligned}$$

This concludes the discussion on the convergence of the variance term.

2.1.4 Bounding the bias term

Exploiting the property that $f \in \Sigma(\beta, L_f)$,

$$\left| \frac{\sum_{i=1}^n (f(X_i) - f(0)) K_h(X_i)}{\sum_{i=1}^n K_h(X_i)} \right| \leq \left| \frac{\sum_{i=1}^n L_f \|X_i\|^\beta K_h(X_i)}{\sum_{i=1}^n K_h(X_i)} \right| \leq L_f h^\beta.$$

Thus, $b^2(0) \leq L_f^2 h^{2\beta}$.

2.1.5 Conclusion

Combining all previous upper bounds yields:

$$\mathbb{E} \left[\left(\hat{f}_n(0) - f(0) \right)^2 \right] \leq n \exp(-Cnh^d) + \frac{C}{nh^d} + Ch^{2\beta}.$$

The first term can be neglected as it is asymptotically dominated by any polynomial decay. Balancing the last two terms yields the optimal bandwidth and the associated risk:

$$\begin{aligned} h_n &\sim n^{-\frac{1}{2\beta+d}}, \\ \mathbb{E} \left[\left(\hat{f}_n(0) - f(0) \right)^2 \right] &\leq Cn^{-\frac{2\beta}{2\beta+d}}. \end{aligned}$$

3 Extension of the result to $\beta \in]1, 2]$

$f \in \Sigma(\beta, L)$ with $\beta \in]1, 2]$ means that $\forall x, y \in \mathbb{R}^d$, $|f(x) - f(y) - \langle \nabla f(y), x - y \rangle| \leq L\|x - y\|^\beta$. In this setting, f must be continuously differentiable. The associated smoothness constraint is rather stringent, as evidenced by these two figures:

Assuming in addition that $p \in \Sigma(\beta - 1, L_p)$, we obtain, for h small enough,

$$b^2(0) \leq \left(L_f + \|\nabla f(0)\| L_p p_{\min}^{-1} \right)^2 h^{2\beta} + \sigma_X^2.$$

Setting

$$G_i = \frac{(f(X_i) - f(0))K_h(X_i)}{\sum_{k=1}^n K_h(X_k)},$$

we obtain the following formulation of $b^2(0)$:

$$b^2(0) = \mathbb{E} \left[\sum_{i,j=1}^n G_i G_j \mathbf{1}(\mathcal{E}) \right].$$

For $i \neq j$, the expectation of the product can be written as:

$$\mathbb{E}[G_i G_j \mathbf{1}(\mathcal{E})] = \mathbb{E}[(f(X_i) - f(0))(f(X_j) - f(0))A(X_i, X_j)]$$

with:

$$A(X_i, X_j) = \frac{K_h(X_i)K_h(X_j)}{(\sum_{k=1}^n K_h(X_k))^2} \mathbf{1}(\mathcal{E}) \geq 0.$$

For conciseness, we omit the explicit dependence of $A(X_i, X_j)$ on the variables $(X_k, k \neq i, k \neq j)$. This leads to:

$$\mathbb{E}'[G_i G_j \mathbf{1}(\mathcal{E})] = \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} (f(x_i) - f(0))(f(x_j) - f(0))A(x_i, x_j) p(x_i)p(x_j) dx_i dx_j$$

where $\mathbb{E}'$ denotes the expectation over (X_i, X_j) , with $(X_k, k \neq i, k \neq j)$ held fixed.

Let us define the first-order Taylor remainder:

$$\begin{aligned} R(x_i) &= f(x_i) - f(0) - \langle \nabla f(0), x_i \rangle, \\ R(x_j) &= f(x_j) - f(0) - \langle \nabla f(0), x_j \rangle. \end{aligned}$$

We have

$$\begin{aligned} (f(x_i) - f(0))(f(x_j) - f(0)) &= (R(x_i) + \langle \nabla f(0), x_i \rangle)(R(x_j) + \langle \nabla f(0), x_j \rangle) \\ &= R(x_i)R(x_j) + R(x_i)\langle \nabla f(0), x_j \rangle + R(x_j)\langle \nabla f(0), x_i \rangle + \langle \nabla f(0), x_i \rangle \langle \nabla f(0), x_j \rangle. \end{aligned}$$

Substituting this expansion into the integral yields three terms, since permuting x_i and x_j maps $R(x_i)\langle \nabla f(0), x_j \rangle$ to $R(x_j)\langle \nabla f(0), x_i \rangle$ while leaving $A(x_i, x_j) p(x_i)p(x_j)$, and hence the integral, unchanged:

$$\begin{aligned} \mathbb{E}'[G_i G_j \mathbf{1}(\mathcal{E})] &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \langle \nabla f(0), x_i \rangle \langle \nabla f(0), x_j \rangle A(x_i, x_j) p(x_i)p(x_j) dx_i dx_j \\ &\quad + 2 \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \langle \nabla f(0), x_i \rangle R(x_j) A(x_i, x_j) p(x_i)p(x_j) dx_i dx_j \\ &\quad + \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} R(x_i)R(x_j) A(x_i, x_j) p(x_i)p(x_j) dx_i dx_j. \end{aligned}$$

Since the inner product is odd and $A(x_i, x_j)$ is even, it follows that:

$$\int_{\mathbb{R}^d} \langle \nabla f(0), x_i \rangle A(x_i, x_j) p(0) dx_i = 0$$

for all x_j , because the integrand is odd with respect to x_i for any fixed $(X_k, k \neq i, k \neq j)$. Applying this observation to x_i and x_j in the first term, as well as in the second term, the decomposition becomes:

$$\begin{aligned} \mathbb{E}'[G_i G_j \mathbf{1}(\mathcal{E})] &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \langle \nabla f(0), x_i \rangle \langle \nabla f(0), x_j \rangle A(x_i, x_j) (p(x_i) - p(0)) (p(x_j) - p(0)) dx_i dx_j \\ &\quad + 2 \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \langle \nabla f(0), x_i \rangle R(x_j) A(x_i, x_j) (p(x_i) - p(0)) p(x_j) dx_i dx_j \\ &\quad + \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} R(x_i) R(x_j) A(x_i, x_j) p(x_i) p(x_j) dx_i dx_j. \end{aligned}$$

The Hölder class condition implies that $|R(x_i)| \leq L_f \|x_i\|^\beta$. Moreover, $A(x_i, x_j)$ vanishes whenever $\|x_i\| > h$ or $\|x_j\| > h$. Applying the Cauchy–Schwarz inequality to the inner products along with the Hölder assumption on p , we obtain the following bound:

$$\begin{aligned} \mathbb{E}'[G_i G_j \mathbf{1}(\mathcal{E})] &\leq B^2 L_p^2 h^{2\beta} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} A(x_i, x_j) dx_i dx_j \\ &\quad + 2 B L_f L_p h^{2\beta} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} A(x_i, x_j) p(x_j) dx_i dx_j \\ &\quad + L_f^2 h^{2\beta} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} A(x_i, x_j) p(x_i) p(x_j) dx_i dx_j \end{aligned}$$

where $B = \|\nabla f(0)\|$.

Factoring out common terms, exploiting the lower bound $p_{\min}$ of the density, and taking the expectation over the remaining variables $(X_k, k \neq i, k \neq j)$, we establish:

$$\mathbb{E}[G_i G_j \mathbf{1}(\mathcal{E})] \leq h^{2\beta} (B L_p p_{\min}^{-1} + L_f)^2 \mathbb{E} \left[\frac{K_h(X_i) K_h(X_j)}{(\sum_{k=1}^n K_h(X_k))^2} \mathbf{1}(\mathcal{E}) \right].$$

When $j = i$, since $K_h(X_1)$ vanishes unless $\|X_1\| \leq h$, we obtain a multiple of the variance term:

$$\sum_{i=1}^n \mathbb{E}[G_i^2 \mathbf{1}(\mathcal{E})] = n \mathbb{E} \left[(f(X_1) - f(0))^2 \frac{K_h(X_1)^2}{(\sum_{k=1}^n K_h(X_k))^2} \mathbf{1}(\mathcal{E}) \right] \leq \sup_{\|u\| \leq h} |f(u) - f(0)|^2 \sigma_X^2,$$

which is at most σ_X^2 as soon as h is small enough, f being continuous at 0.

Finally, for the cross-product terms:

$$\begin{aligned} \sum_{i \neq j} \mathbb{E}[G_i G_j \mathbf{1}(\mathcal{E})] &\leq h^{2\beta} (B L_p p_{\min}^{-1} + L_f)^2 \mathbb{E} \left[\frac{\sum_{i \neq j} K_h(X_i) K_h(X_j)}{(\sum_{k=1}^n K_h(X_k))^2} \mathbf{1}(\mathcal{E}) \right] \\ &\leq h^{2\beta} (B L_p p_{\min}^{-1} + L_f)^2 \mathbb{E} \left[\frac{\sum_{i,j=1}^n K_h(X_i) K_h(X_j)}{(\sum_{k=1}^n K_h(X_k))^2} \mathbf{1}(\mathcal{E}) \right] \\ &\leq h^{2\beta} (B L_p p_{\min}^{-1} + L_f)^2. \end{aligned}$$

4 Explanation of the breakthrough

This result, counter-intuitive in light of classical intuition, can be explained by two main arguments.

Interpolation has historically raised concerns regarding the inflation of the variance term. Returning to inequality (1), the two arguments proceed as follows.

The first argument applies to the second term. Assuming $\alpha < d/2$, the kernel is square-integrable (L^2). In terms of this norm, the kernel values around the singularity are sufficiently small so that, asymptotically, the effect of interpolation around this point becomes increasingly localized. As more kernels are aggregated in this model, their individual influence diminishes (cf. Figure 2).

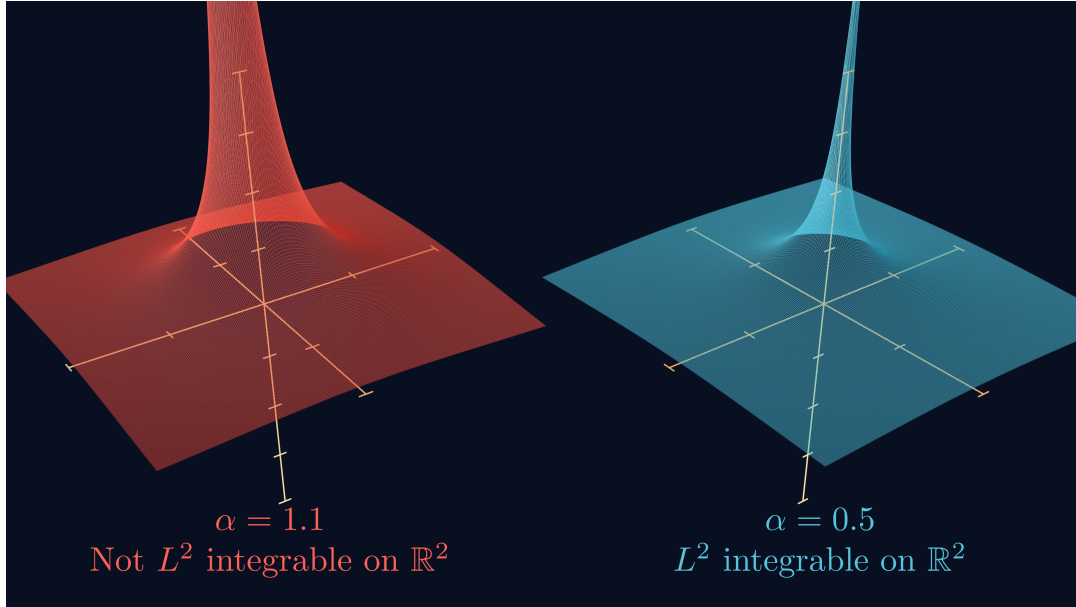


Figure 2: Localization of the kernel around the singularity in the case $\alpha < d/2$.

The second argument consists in upper bounding the probability that the origin 0 lies abnormally far from too many sample points $(X_i)_{i=1}^n$. Since this event is exponentially unlikely (by Bernstein’s inequality), one can deduce that only the behavior of the estimator when 0 is sufficiently close to the sample points X_i governs the quality of the estimation (the phrase *sufficiently close* meaning that 0 is within distance less than h of at least $n\bar{p}/2$ distinct points X_i).

5 Simulations

In this section, we illustrate the empirical performance of the singular Nadaraya–Watson estimator via numerical simulations on a regression and a binary classification task.

5.1 Regression

By choosing $f(x) = x^2$ as the function to estimate on the interval $[0, 1]$ we can see illustrated in Figure 3 the ability of the estimator to interpolate data while keeping the overfitting sufficiently localized for the model to asymptotically generalize at an optimal speed. The parameter α has been set to 0.49 throughout the simulations as $d = 1$ to keep the relation $0 < \alpha < d/2$ true.

In Figure 4, the mean squared error has been computed for a generator defined for datasets with a size starting from 10 to 1000 points. The errors have been averaged over 20 iterations to approximate the associated risk as closely as possible. The function f being Hölder on $[0, 1]$ with a value of β equal to 1, the convergence speed is in $O\left(n^{-\frac{2}{3}}\right)$. With the help of a polynomial regression, a curve has been fitted and shows how accurate the inequality is.



Figure 3: Illustration of the properties of the singular N–W estimator.

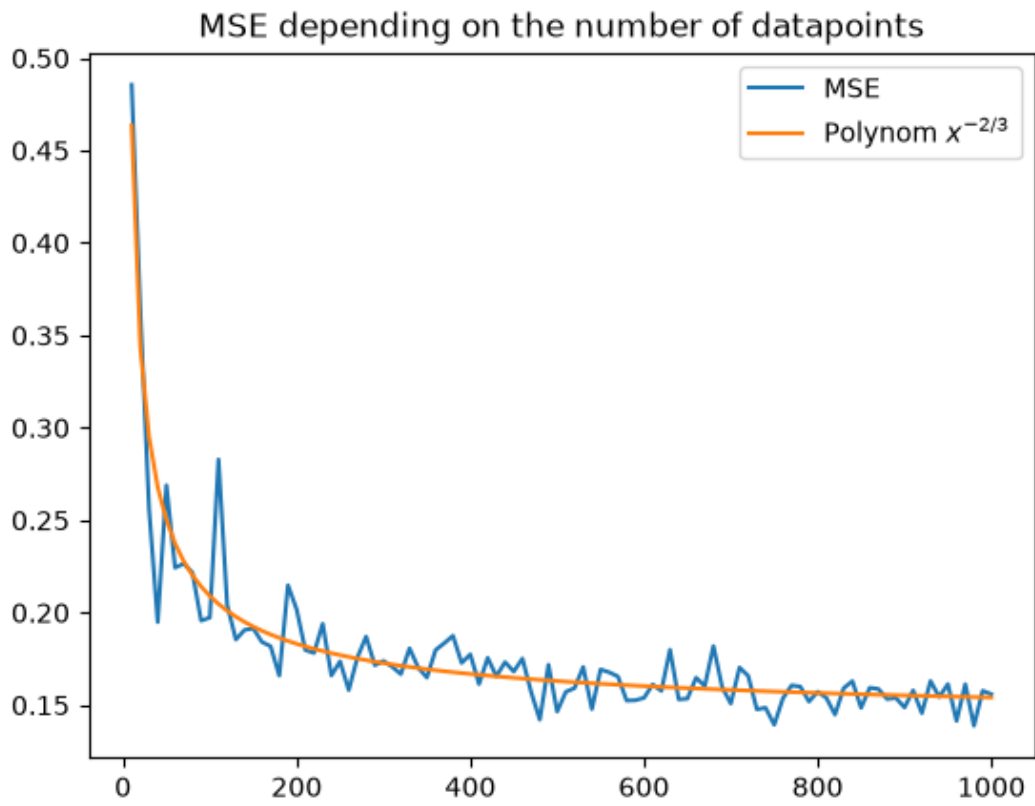


Figure 4: Convergence of the MSE for $f(x) = x^2$.

5.2 Classification

5.2.1 Experimental setting

We consider a binary classification setting where the target variable Y takes values in $\{-1, 1\}$. We investigate the behavior of the estimator under two distinct configurations for the distribution of the covariates X :

1. **Gaussian model:** The class-conditional densities, denoted by $f_{Y=-1}(X)$ and $f_{Y=1}(X)$, are Gaussian densities with respective means $\mu_{-1} = -(0.5, 0.5)$ and $\mu_1 = (0.5, 0.5)$. They share a common covariance matrix Σ . To evaluate the robustness of the model to label noise and class overlap, three covariance regimes are evaluated:
 - $\Sigma = \text{Id}$ (baseline model),
 - $\Sigma = 1.2 \cdot \text{Id}$ (intermediate variance),
 - $\Sigma = 2 \cdot \text{Id}$ (high variance).
2. **Nonlinear model:** To probe the adaptability of the method when faced with geometrically more complex decision boundaries, a nonlinear setting is considered. The conditional probability is defined by:

$$\mathbb{P}(Y = -1 \mid X) = f(X) \quad \text{with} \quad f(X) = \frac{-\frac{X_1^2}{16} + 0.5}{0.833}.$$

5.2.2 Classification method

To separate the two classes, the Bayes decision rule consists in evaluating the sign of the regression function $\mathbb{E}[Y \mid X]$. In practice, we evaluate the singular Nadaraya–Watson estimator constructed using a kernel that exhibits a singularity at the origin. The predicted label $\hat{Y}$ at a query point x is directly obtained by thresholding the estimator:

$$\hat{Y} = \text{sign} \left(\hat{f}_n(x) \right).$$

Although the estimator strictly interpolates the training sample, tuning the bandwidth alongside a proper singularity exponent $\alpha < d/2$ ensures that the variance remains asymptotically controlled, thereby yielding a stable decision boundary at the macro scale.

5.2.3 Results and analysis

Figures 5, 6, and 7 depict the decision boundaries obtained for the Gaussian model across the three variance settings under study.

Inspecting these initial figures indicates that despite increasing class overlap as the variance grows, the estimator preserves a globally consistent separation. The interpolation induced by the kernel singularity acts locally around the training samples without triggering macroscopic oscillations or catastrophic overfitting, visually validating the theoretical risk bounds.

Lastly, Figure 8 displays the classification boundaries obtained under the nonlinear scenario.

In this setting, the resulting decision boundary deviates locally from the theoretical parabolic expectation. This highlights the capacity of the singular Nadaraya–Watson estimator to adapt to complex geometric structures through local flexibility while preventing global overfitting.

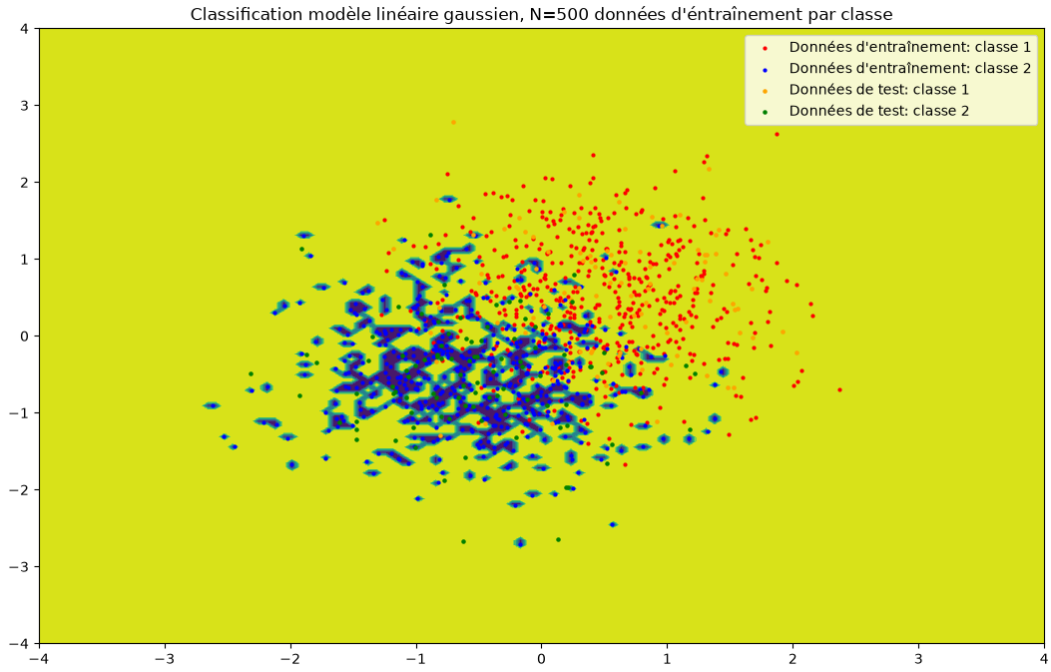


Figure 5: Classification under the Gaussian model with baseline variance ($\mu_1 = (0.5, 0.5)$, $\mu_{-1} = -(0.5, 0.5)$, $\Sigma = \text{Id}$).

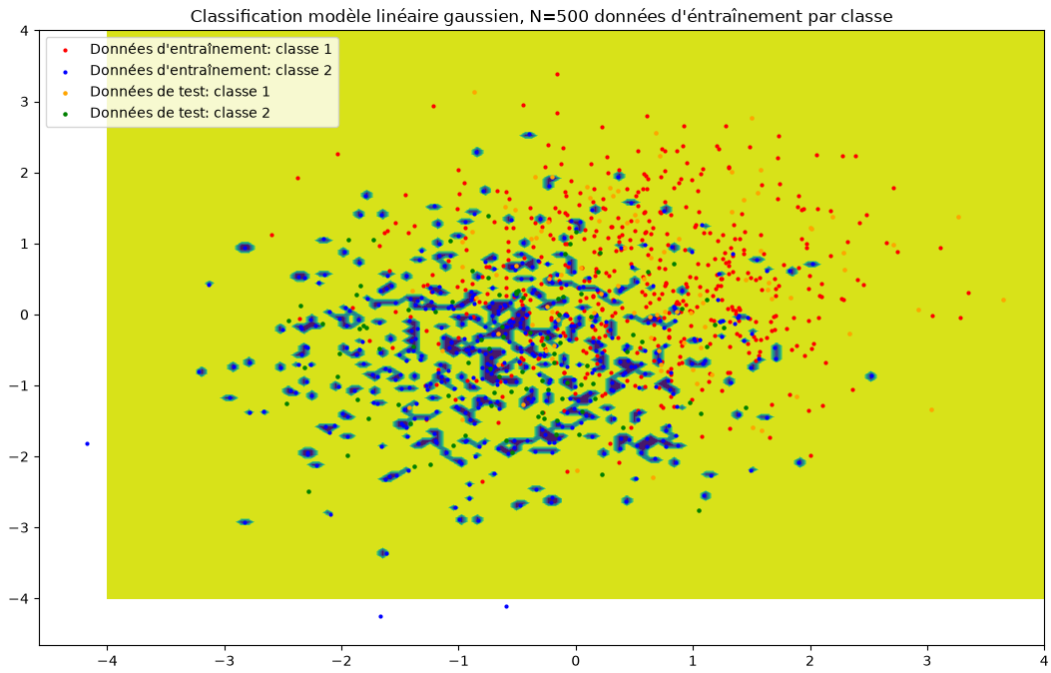


Figure 6: Classification under the Gaussian model with intermediate variance ($\mu_1 = (0.5, 0.5)$, $\mu_{-1} = -(0.5, 0.5)$, $\Sigma = 1.2 \cdot \text{Id}$).

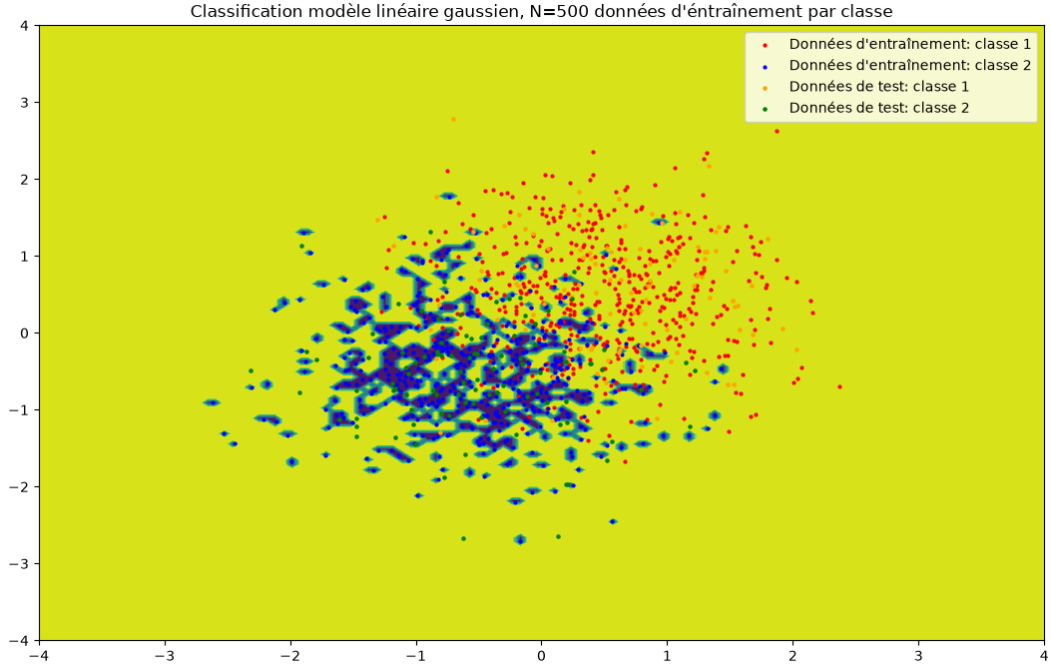


Figure 7: Classification under the Gaussian model with high variance ($\mu_1 = (0.5, 0.5)$, $\mu_{-1} = -(0.5, 0.5)$, $\Sigma = 2 \cdot \text{Id}$).

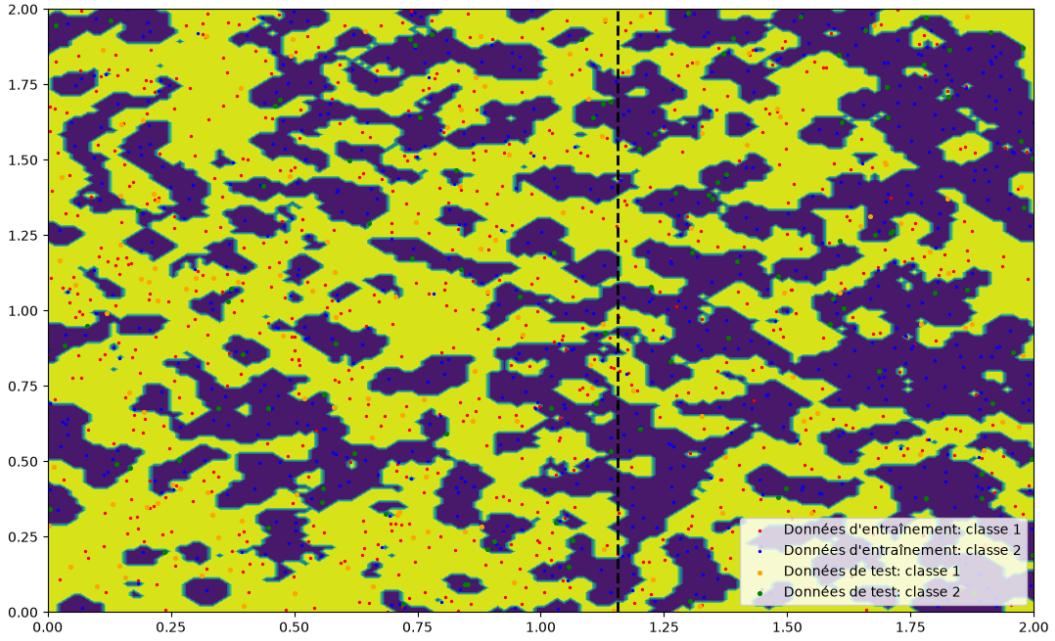


Figure 8: Classification under the nonlinear model with conditional distribution $\mathbb{P}(Y = -1 \mid X) = \frac{-X_1^2/16+0.5}{0.833}$.

References

- [1] M. Belkin, A. Rakhlin, and A. B. Tsybakov. Does data interpolation contradict statistical optimality? In *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 89 of *Proceedings of Machine Learning Research*, pages 1611–1619. PMLR, 2019. [arXiv:1806.09471](#).